Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Length of a latus rectum of the ellipse $\frac{x^2}{81} + \frac{y^2}{63} = 1$ is (in units)

(A) 9

(B) 7

(C) 14

(D) 18

Q.2

The locus of the point which divides the double ordinates of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the ratio $1:2$ internally is

(A) $\frac{x^2}{a^2} + \frac{9y^2}{b^2} = 1$

(B) $\frac{x^2}{a^2} + \frac{9y^2}{b^2} = \frac{1}{9}$

(C) $\frac{9x^2}{a^2} + \frac{9y^2}{b^2} = 1$

(D) none of these

Q.3

The normal at the point $(3, 2\sqrt{3})$ on the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ meets the major axis of the ellipse at $(\alpha, 0)$, the value of α is:

(A) $\frac{5}{3}$

(B) $\frac{3}{5}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Q.4

The tangent to the ellipse $3x^2 + 16y^2 = 12$, at the point $(1, 3/4)$, intersects the curve $y^2 + x = 0$ at:

(A) no point

(B) exactly one point

(C) two distinct points

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(D) more than two points

Q.5

The point of intersection of the tangents at the point P on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and its corresponding point Q on the auxiliary circle meet on the line

(A) $x=a/e$

(B) $x=0$

(C) $y=0$

(D) none of these

Q.6

P and Q are the foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and B is an end of the minor axis. If PBQ is an equilateral triangle, then the eccentricity of the ellipse is

(A) $1/\sqrt{2}$

(B) $1/3$

(C) $1/2$

(D) $\sqrt{3}/2$

Q.7

If a and b are the natural numbers such that $a+b=ab$, then equation of the chord of the ellipse $x^2+4y^2=4$ with (a,b) as the mid point is:

(A) $x+4y=2$

(B) $x+4y=10$

(C) $4x+y=2$

(D) $4x+y=10$

Q.8

P is a point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b)$ and Q is the point corresponding to P on the auxiliary circle $x^2+y^2=a^2$.N is the foot of the perpendicular from P on the major axis of the ellipse. $\frac{PN}{PQ}$ is equal to:

(A) $\frac{b}{a-b}$

(B) $\frac{a}{a-b}$

(C) $\frac{b}{a+b}$

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(D) $\frac{a}{a+b}$

Q.9

$C_1 : x^2 + y^2 = a^2$ and $C_2 : x^2 + y^2 = b^2$ are two contentric circles. A line through the centre of these circles intersects them at P and Q respectively. If the lines through P and Q parallel to y-axis and x-axis respectively meet at the point R, then the locus of R is

(A) $x^2 + y^2 = a^2 + b^2$

(B) $x^2 - y^2 = a^2 - b^2$

(C) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(D) $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

Q.10

The distance of the point $(\sqrt{6}\cos\theta, \sqrt{2}\sin\theta)$ on the ellipse from the centre of the ellipse is 2 if $\theta =$

(A) $\pi/6$

(B) $\pi/4$

(C) $\pi/3$

(D) none of these

Q.11

The number of distinct normal lines that can be drawn to the ellipse $\frac{x^2}{169} + \frac{y^2}{25} = 1$ from the point $P(0, 6)$ is

(A) one

(B) two

(C) three

(D) four

Q.12

If the normal at any point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the axis of x at G and the axis of y at g, then $PG : pg$ is:

(A) $a^2 : b^2$

(B) $a : b$

(C) $b : a$

(D) $b^2 : a^2$

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Q.13

Equation of an ellipse with centre at the origin passing through $(5, 0)$ and having eccentricity $2/3$ is:

(A) $4x^2 + 9y^2 = 100$

(B) $9x^2 + 5y^2 = 225$

(C) $5x^2 + 9y^2 = 125$

(D) $6x^2 + 4y^2 = 150$

Q.14

If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then the maximum value of ab is

(A) 4

(B) 2

(C) 1

(D) none of these

Q.15

If chords of contact of the tangent from two points (x_1, y_1) and (x_2, y_2) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are at right angles, then $\frac{x_1 x_2}{y_1 y_2}$ is equal to

(A) $\frac{a^2}{b^2}$

(B) $\frac{-b^2}{a^2}$

(C) $\frac{-a^4}{b^4}$

(D) $\frac{b^4}{a^4}$

Q.16

The locus of the point of intersection of the tangent at the endpoints of the focal chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. (b < a)$ is a/an

(A) circle

(B) ellipse

(C) hyperbola

(D) pair of straight lines

Q.17

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The area of the quadrilateral formed by the tangents at the endpoint of the latus rectum to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ is

(A) $27/4$ sq. units

(B) 9 sq. units

(C) $27/2$ sq. units

(D) 27 sq. units

Q.18

The eccentricity of the locus of point $(3h+2, k)$, where (h, k) lies on the circle $x^2+y^2=1$, is

(A) $1/3$

(B) $\sqrt{2}/3$

(C) $2\sqrt{2}/3$

(D) $1/\sqrt{3}$

Q.19

$P_1\left(\theta_1\right)$ and $P_2\left(\theta_2\right)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ such that $\tan\theta_1 \tan\theta_2=\frac{-a^2}{b^2}$. The chord joining P_1 and P_2 of the ellipse subtends a right angle at the

(A) focus $(ae, 0)$

(B) focus $(-ae, 0)$

(C) centre $(0, 0)$

(D) vertex $(a, 0)$

Q.20

If the tangent at a point on the ellipse $\frac{x^2}{27}+\frac{y^2}{3}=1$ meets the coordinate axes at A and B, and O is the origin, then the minimum area (in sq. units) of the triangle OAB is

(A) $3\sqrt{3}$

(B) $\frac{9}{2}$

(C) 9

(D) $9\sqrt{3}$

Q.21

If the line $y=mx+c$ is a tangent to the ellipse $x^2+2y^2=4$ then c cannot take the value

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Q.22

Let the distance between a focus and the corresponding directrix of an ellipse be 8 and the eccentricity be $1/2$. If the length of the minor axis is k, then $\sqrt{3}k/2$ is

Q.23

Square of the radius of director circle of auxiliary circle of ellipse $(3x+4y-1)^2+5(4x-3y+2)^2=250$ is

Q.24

The number of points on the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ from which a pair of perpendicular tangents is drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ is

Q.25

If CF is perpendicular from the centre of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ to the tangent at P, G is the point where the normal at P meets the major axis, then $CF \cdot PG =$

Q.26

If the variable line $y=kx+2h$ is tangent to an ellipse $2x^2+3y^2=6$, then the locus of $P(h,k)$ is a conic C whose eccentricity is e. Then the value of $3e^2$ is

Q.27

Points P and D are taken on the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$. If a b, c and d are the lengths of the sides of quadrilateral $PADB$, where A and B are foci of the ellipse, then maximum value of $(abcd)$ is

Q.28

If the midpoint of a chord of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is $(0,3)$ then the length of the chord is (mark answer to nearest integer)

Q.29

Tangents drawn from the point $P(2,3)$ to the circle $x^2+y^2-8x+6y+1=0$ touch circle at points A and B. The circumcircle of ΔPAB cuts the director circle of ellipse $\frac{(x+5)^2}{9}+\frac{(y-3)^2}{b^2}=1$ orthogonally. Then the value of b^2 is

Q.30

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For the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with vertices A and A', tangent drawn at the point P in the first quadrant meets the y-axis at Q and the chord $A'P$ meets the y-axis at M. If O is the origin, then OQ^2-MQ^2 is equal to

Answers & Solutions

Q.1 Answer:

14

Solution:

Length of the latus rectum of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{2b^2}{a}$ so the required length is $\frac{2\times 63}{9}=14$ units

Q.2 Answer:

$\frac{x^2}{a^2}+\frac{9y^2}{b^2}=1$

Solution:

R divides the line PQ in the ratio $1:2$ then,

$$\mathrm{h}=\frac{1\times \mathrm{a}\cos\theta+2\times \mathrm{a}\cos\theta}{3}$$

$$\Rightarrow \mathrm{h}=\mathrm{a}\cos\theta$$

$$\Rightarrow \frac{\mathrm{h}}{\mathrm{a}}=\cos\theta$$

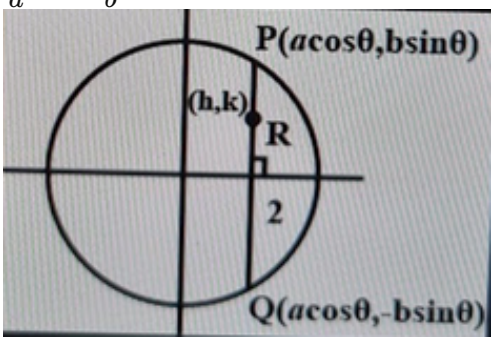
And, $\mathrm{k}=\frac{1\times(-\mathrm{b}\sin\theta)+2\times \mathrm{b}\sin\theta}{3}$

$$\mathrm{k}=\frac{\mathrm{b}\sin\theta}{3}$$

$$\Rightarrow \frac{3\mathrm{k}}{\mathrm{b}}=\sin\theta$$

Squaring and adding equations 1 and 2 we get:

$\frac{x^2}{a^2}+\frac{9y^2}{b^2}=1$



Q.3 Answer:

$\frac{5}{3}$

Solution:

Equation of the normal is

$$\frac{x-3}{3}=\frac{\frac{y-2\sqrt{3}}{2\sqrt{3}}}{16}$$

$$\Rightarrow 12\sqrt{3}(x-3)=8(y-2\sqrt{3})$$

which meets the major axis $y=0$ at point for which

$$12\sqrt{3}(x-3)=-16\sqrt{3}\Rightarrow x=\frac{5}{3}$$

$$\Rightarrow \alpha=\frac{5}{3}$$

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Q.4 Answer:

exactly one point

Solution:

Tangent at $\left(1, \frac{3}{4}\right)$ **to the ellipse** $3x^2 + 16y^2 = 12$ **is**

$\frac{x \cdot 1}{4} + \frac{y\left(\frac{3}{4}\right)}{\frac{3}{4}} = 1$

$\Rightarrow x + 4y = 4$

which intersects the curve $y^2 + x = 0$ **at points for which**

$y^2 + (4 - 4y) = 0 \Rightarrow (y-2)^2 = 0 \Rightarrow y = 2$

and the points of intersection is $(-4, 2)$ **exactly one point.**

Q.5 Answer:

$y = 0$

Solution:

$P(a\cos\theta, b\sin\theta) \quad L_P : \quad \frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

$Q(a\cos\theta, a\sin\theta) \quad L_Q : \quad x\cos\theta + y\sin\theta = a$

$L_P - \frac{L_Q}{a}$

$\Rightarrow \left(\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta - 1\right) - \left(\frac{x}{a}\cos\theta + \frac{y}{a}\sin\theta - 1\right) = 0$

$y\left(\frac{\sin\theta}{b} - \frac{\sin\theta}{a}\right) = 0$

$y = 0$

Q.6 Answer:

$1/2$

Solution:

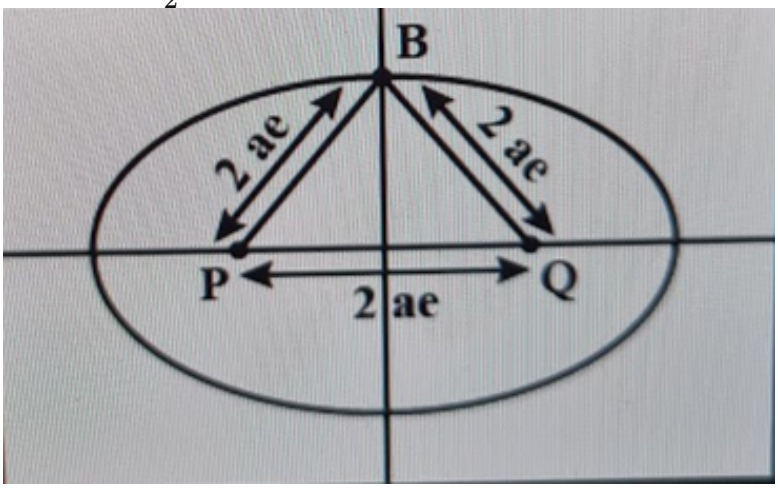
Since the $\triangle$PBQ **is equilateral then,**

All sides are 2ae

And, $\mathrm{PB} + \mathrm{BQ} = 2\mathrm{a}$

$\Rightarrow 4ae = 2a$

$\Rightarrow e = \frac{1}{2}$



Q.7 Answer:

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$x + 4y = 10$

Solution:

Since a and b are natural numbers such that $a + b = ab, a = b = 2$

Equation of the chord of the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ having (a, b) as the mid point is $\frac{x(2)}{4} + \frac{y(2)}{1} - 1 =$

$\frac{2^2}{4} + \frac{2^2}{1} - 1\,(T = S_1)$

$\Rightarrow x + 4y = 10$

Q.8 Answer:

$\frac{b}{a-b}$

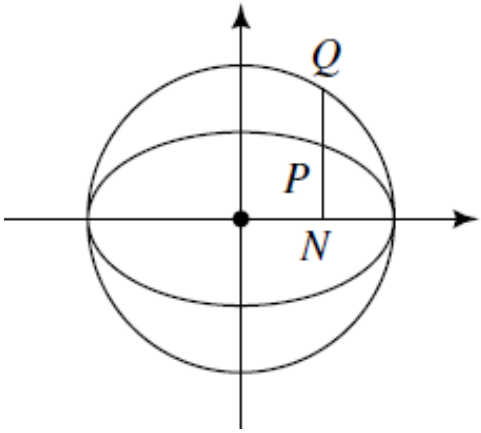
Solution:

Let the coordinates of P be $(a\cos\theta, b\sin\theta)$ then coordinate of Q are $(a\cos\theta, a\sin\theta)$

$PN = b\sin\theta, QN = a\sin\theta$

$\Rightarrow PQ = (a - b)\sin\theta$

and $\frac{PN}{PQ} = \frac{b}{a-b}$.

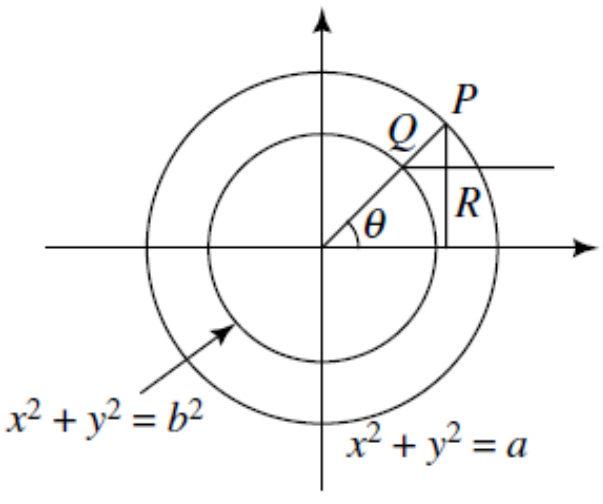


Q.9 Answer:

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

Coordinates of P are $(a\cos\theta, a\sin\theta)$ and of Q are $(b\cos\theta, b\sin\theta)$. Equation of the line through P parallel to y-axis is $x = a\cos\theta$ and of the line through Q parallel to x-axis is $y = b\sin\theta$. So the locus of the point of intersection is $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = \cos^2\theta + \sin^2\theta = 1$



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Q.10 Answer:

$\pi/4$

Solution:

$6\cos^2\theta + 2\sin^2\theta = 4 \Rightarrow \cos^2\theta = 1/2$ **so** $\theta = \pi/4$

Q.11 Answer:

three

Solution:

Given $\frac{x^2}{160} + \frac{y^2}{25} = 1$

$\Rightarrow a = 13, \quad b = 5$

Now normal to the cllipse at $(13\cos\theta, 5\sin\theta)$**is** $\frac{13x}{\cos\theta} - \frac{5x}{\sin\theta} = 169 - 25 = 144$

This passes throngh point $(0, 6)$ **so,**

$\frac{-30}{\sin\theta} = 144$

$\sin\theta = \frac{-15}{72} = \frac{-5}{24}$

$\Rightarrow \quad \theta = \pi + \sin^{-1}\left(\frac{5}{24}\right)$

Hence the no of normals that can pass through $(0, 6)$ **is 3**

Q.12 Answer:

$b^2 : a^2$

Solution:

Equation of the normal at $P(a\cos\theta, b\sin\theta)$ **on the ellipse is** $ax\sec\theta - by\,\mathrm{cosec}\,\theta = a^2 - b^2$

So coordinates of G **are** $\left(\frac{a^2-b^2}{a\sec\theta}, 0\right)$ **and of** g **are**

$\left(0, -\frac{a^2-b^2}{a\,\mathrm{cosec}\,\theta}\right)$

$(PG)^2 = \left(a\cos\theta - \frac{a^2-b^2}{a}\cos\theta\right)^2 + b^2\sin^2\theta$

$= \frac{b^2}{a^2}\left(b^2\cos^2\theta + a^2\sin^2\theta\right)$

and $(Pg)^2 = \frac{a^2}{b^2}\left(b^2\cos^2\theta + a^2\sin^2\theta\right)$

So $PG : Pg = b^2 : a^2$

Q.13 Answer:

$5x^2 + 9y^2 = 125$

Solution:

Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$**. It passes through** $(5, 0) \Rightarrow a^2 = 25$

$b^2 = a^2\left(1 - e^2\right) \Rightarrow b^2 = 25\left(1 - \frac{4}{9}\right) = \frac{125}{9}$

Hence the required equation of the ellipse is $5x^2 + 9y^2 = 125$

Q.14 Answer:

4

Solution:

A tangent of slope 2 to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ **is**

$y=2x\pm\sqrt{4a^2+b^2}$

This is normal to the circle $x^2+y^2+4x+1=0$

$\Rightarrow$ **Equation (i), passes through center of the circle** $(-2,0)$

$\Rightarrow 0=-4\pm\sqrt{4a^2+b^2}$

$\Rightarrow 4a^2+b^2=16$

Usin A.M $\geq$ **G.M., we get**

$\frac{4a^2+b^2}{2}\geq\sqrt{4a^2b^2}$

$\Rightarrow ab\leq 4$

$\therefore$ **the minimum value of** ab **is 4**

Q.15 Answer:

$\frac{-a^4}{b^4}$

Solution:

Chords of contacts are

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ and $\frac{xx_2}{a^2}+\frac{yy_2}{b^2}=1$

Since they are at right angles

$\frac{-b^2}{a^2}\times\frac{x_2}{y_2}\times\frac{-b^2}{a^2}\times\frac{x_1}{y_1}=-1$

$\Rightarrow\frac{x_1x_2}{y_1y_2}=\frac{-a^4}{b^4}$

Q.16 Answer:

pair of straight lines

Solution:

By the property of ellipse, That if two tangents interests at focal cord they meet at that point on the ellipse

so,

$x=\pm\frac{a}{e}$

$=\pm\frac{a}{\sqrt{1-\frac{b^2}{a^2}}}$

$x=\pm\frac{a^2}{\sqrt{a^2-b^2}}$

Q.17 Answer:

27 sq. units

Solution:

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Given equation of ellipse is

$\frac{x^2}{9}+\frac{y^2}{5}=1 \quad \ldots\ldots(1)$

$\therefore a^2=9, b^2=5$

$\Rightarrow a=3, b=\sqrt{5}$

Now, $e=\sqrt{1-\frac{b^2}{a^2}}=\sqrt{1-\frac{5}{9}}=\frac{2}{3}$

Foci $=(\pm ae, 0)=(\pm 2,0)$

and $\frac{b^2}{a}=\frac{5}{3}$

$\therefore$ Extremities of one of latus rectum are $\left(2, \frac{5}{3}\right)$ and $\left(2, \frac{-5}{3}\right)$

$\therefore$ Equation of tangent at $\left(2, \frac{5}{3}\right)$ is

$\frac{x(2)}{9}+\frac{y\left(\frac{5}{3}\right)}{5}=1$ or $2x+3y=9$

$\mathrm{Eqn}(2)$ intersects X and Y axes at $\left(\frac{9}{2}, 0\right)$ and $(0,3)$ respectively.

$\therefore$ Area of quadrilateral $4\times$ Area of $\triangle \mathrm{POQ}$

$=4 \times \frac{1}{2} \times \frac{9}{2} \times 3=27$ sq.units.

Q.18 Answer:

$2\sqrt{2}/3$

Solution:

Let $\mathrm{p}=3\,\mathrm{h}+2$ and $\mathrm{q}=\mathrm{k}$

$\Rightarrow \mathrm{h}=\frac{\mathrm{p}-2}{3}$ and $\mathrm{k}=\mathrm{q}$

Since (h, k) lies on $\mathrm{x}^2+\mathrm{y}^2=1$

$\Rightarrow \mathrm{h}^2+\mathrm{k}^2=1$

$\Rightarrow \left(\frac{\mathrm{p}-2}{3}\right)^2+\mathrm{q}^2=1$

Locus is $\left(\frac{x-2}{3}\right)^2+y^2=1$

Which has eccentricity $\mathrm{e}=\sqrt{1-\frac{1}{9}}=\frac{2\sqrt{2}}{3}$

Q.19 Answer:

centre $(0,0)$

Solution:

$P_1\left(a\cos\theta_1, b\sin\theta_1\right), P_2\left(a\cos\theta_2, b\sin\theta_2\right)$

Slope of $OP_1=\frac{b\sin\theta_1}{a\cos\theta_1}=\frac{b}{a}\tan\theta_1$

Slope of $OP_2=\frac{b\sin\theta_2}{a\cos\theta_2}=\frac{b}{a}\tan\theta_2$

Product of the slope $=\frac{b^2}{a^2}\tan\theta_1\tan\theta_2$

$=\frac{b^2}{a^2}\left(\frac{-a^2}{b^2}\right)=-1$

So PQ subtends a right angle at $(0,0)$, the centre.

Q.20 Answer:

9

Solution:

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An equation of tangent at $(3\sqrt{3}\cos\theta, \sqrt{3}\sin\theta)$ to the ellipse $\frac{1}{27}x^2 + \frac{1}{3}y^2 = 1$ is $x\cos\theta + 3y\sin\theta = 3\sqrt{3}$

It meets the axes in $A(3\sqrt{3}\sec\theta, 0)$, and $B(0, \sqrt{3}\operatorname{cosec}\theta)$

Let $A =$ area of ΔOAB is $\frac{1}{2}|(3\sqrt{3}\sec\theta)(\sqrt{3}\operatorname{cosec}\theta)|$

$= \frac{9}{|\sin 2\theta|} \geq 9$

Thus, least value of A is 9 which is attained when $\theta = \pi/4, 3\pi/4, 5\pi/4$, or $7\pi/4$

Q.21 Answer:

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Solution:

Using condition of tangency, we get

$c^2 = 4m^2 + 2 \geq 2 \forall m \in R$

$\therefore c^2 \geq 2$

$\Rightarrow |c| \geq \sqrt{3}$

Q.22 Answer:

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Solution:

Distance of directrix from centre $= \frac{a}{c}$ we have

$\mathrm{e} = \frac{1}{2}$

$\frac{a}{e} - \mathrm{ae} = 8$

$\Rightarrow 2\mathrm{a} - \frac{\mathrm{a}}{2} = 8$

$\therefore \mathrm{a} = \frac{16}{3}$

$\mathrm{e} = \sqrt{1 - \frac{\mathrm{b}^2}{\mathrm{a}^2}}$

$= \frac{1}{2} = \sqrt{1 - \frac{\mathrm{b}^2}{\frac{256}{9}}} \& \ \mathrm{b} = \frac{8}{\sqrt{3}}$

$\therefore 2\,\mathrm{b} = \frac{16}{\sqrt{3}}$

Q.23 Answer:

—

Solution:

We have ellipse,

$\frac{\left(\frac{3x+4y-1}{5}\right)^2}{10} + \frac{\left(\frac{4x-3y+2}{5}\right)^2}{2} = 1$

Here, $a^2 = 10$ and $b^2 = 2$

So, radius of the auxiliary circle is $\sqrt{10}$

Hence, radius of director circel of auxiliary circle $\sqrt{2} \times \sqrt{10} = \sqrt{20}$

Q.24 Answer:

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Solution:

For the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, equation of director circle is $x^2+y^2=25$. The director circle will cut the ellipse $\frac{x^2}{50}+\frac{y^2}{20}=1$ at 4 points

Hence, the number of points $=4$.

Q.25 Answer:

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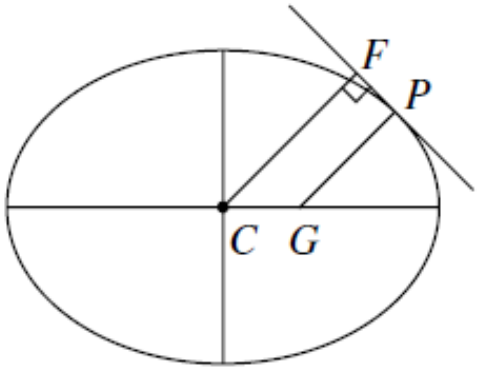
Solution:

Let $P(5\cos\theta, 3\sin\theta)$

Equation of the tangent at P is

$\frac{x\cos\theta}{5}+\frac{y\sin\theta}{3}=1$

$\Rightarrow CF=\frac{1}{\sqrt{\frac{\cos^2\theta}{25}+\frac{\sin^2\theta}{9}}}=\frac{15}{\sqrt{9\cos^2\theta+25\sin^2\theta}}$



Equation of the normal at P is

$\frac{5x}{\cos\theta}-\frac{3y}{\sin\theta}=25-9=16$

Coordinates of G are $\left(\frac{16\cos\theta}{5}, 0\right)$

$\Rightarrow PG=\frac{3}{5}\sqrt{9\cos^2\theta+25\sin^2\theta}$

Hence $CF\cdot PG=9$

Q.26 Answer:

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Solution:

Line $y=kx+2h---(i)$

Slope $m=k$ and constant $c_1=2h$

Ellipse: $2x^2+3y^2=6$

$\Rightarrow \frac{x^2}{3}+\frac{y^2}{2}=1---(ii)$

If (i) is tangent to (ii)

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$\Rightarrow c_1 = \sqrt{a^2m^2 + b^2}$

$\Rightarrow 4h^2 = 3k^2 + 2$

$\Rightarrow 4h^2 - 3k^2 = 2$

$\Rightarrow \frac{h^2}{\frac{1}{2}} - \frac{k^2}{\frac{2}{3}} = 1$

For locus of $\mathrm{P(h,k)}$**, replacing** $\mathrm{h \to x, k \to y}$

$\Rightarrow \frac{\mathrm{x}^2}{\frac{1}{2}} - \frac{\mathrm{y}^2}{2} = 1$ is a hyperbola(conic)

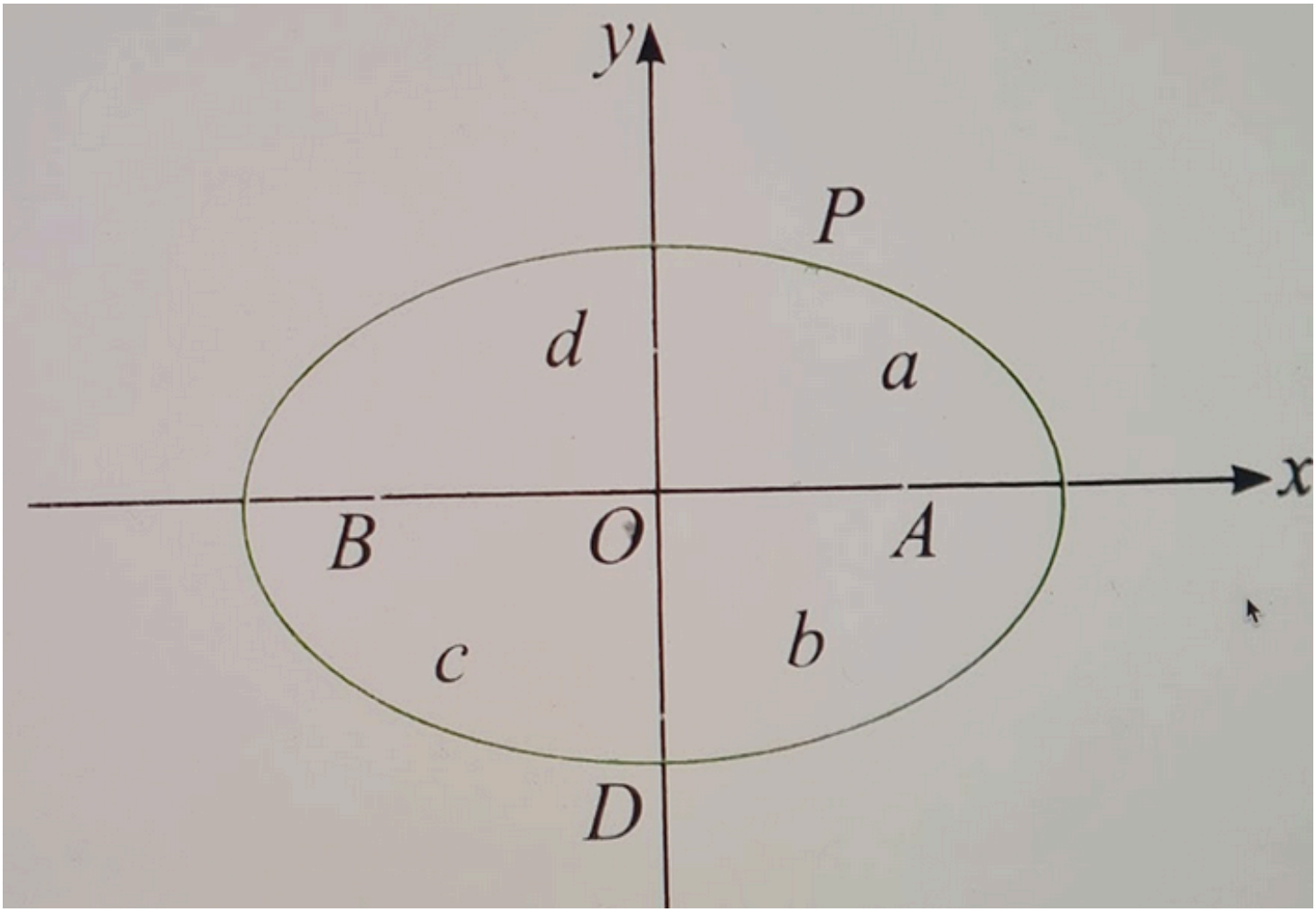
so, eccentricity of this conic $\mathrm{e} = \sqrt{1 + \frac{\left(\frac{2}{3}\right)}{\left(\frac{1}{2}\right)}} \Rightarrow \mathrm{e} = \sqrt{\frac{7}{3}}$

$3\mathrm{e}^2 = \frac{7}{3} \cdot 3 = 7$

Q.27 Answer:

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Solution:



$PA + PB = 4$ **(focal property)**

$\mathrm{DA + DB = 4}$

$\Rightarrow a + b + c + d = 8$

Now, A. $M. \geq G.M$

$\Rightarrow \frac{a+b+c+d}{4} \geq \frac{(abcd)^1}{4}$

$\Rightarrow abcd \leq 16$

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Q.28 Answer:

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Solution:

Ellipse: $\frac{x^2}{16}+\frac{y^2}{25}=1---(i)$

midpoint of chord: $(0,3)$

now equation of chord whose midpoint is given $\Rightarrow S_1=S_{11}$

$\Rightarrow \frac{0(x)}{16}+\frac{3y}{25}-1=\frac{9}{25}-1$

$y=3---(ii)$

now point of intersection of i and ii,

$\text{At } y=3, \frac{x^2}{16}+\frac{9}{25}=1$

$\Rightarrow x^2=\frac{(16)^2}{25}$

$\Rightarrow x=\pm\frac{16}{5}$

now, distance between two points $\left(\frac{16}{5},3\right)$ **and** $\left(\frac{-16}{5},3\right)$

$$\frac{4k}{5}=\sqrt{(3-3)^2+\left[\frac{16}{5}-\left(-\frac{16}{5}\right)\right]^2}$$

$$\frac{4k}{5}=\sqrt{\left(\frac{2(16)}{5}\right)^2}$$

$$\frac{4k}{5}=\frac{32}{5}$$

Q.29 Answer:

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Solution:

Given circle is $x^2+y^2-8x+6y+1=0$. **Center of the given circle is** $O(4,-3)$

OAP B is a cyclic quadrilateral

equation of circumcircle of $\triangle PAB$ **is** $(x-2)(x-4)+(y-3)(y+3)=0$

$x^2+y^2-6x-1=0---(i)$

Directrix of circle of the ellipse: $\frac{(x+5)^2}{9}+\frac{(y-3)^2}{b^2}=1$ **is** $(x+5)^2+(y-3)^2=9+b^2$

$\Rightarrow x^2+y^2+10x-6y+25-b^2=0---(\text{ ii })$

Apply orthogonality property,

$\Rightarrow 2[-3\times5+0\times(-3)]=-1+25-b^2$

$\Rightarrow b^2=54$

Q.30 Answer:

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Solution:

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Let point P be $(\mathrm{a}\cos\theta, \mathrm{b}\sin\theta)$

Equation of the tangent at point P is

$\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1$

Then point Q is $(\mathrm{b}\,\mathrm{cosec}\,\theta, 0)$

Equation of chord A'P is

$\mathrm{y} - 0 = \frac{\mathrm{b}\sin\theta}{\mathrm{a}\cos\theta + \mathrm{a}}(\mathrm{x} + \mathrm{a})$

Putting $x = 0$, we have $y = \frac{b\sin\theta}{\cos\theta+1}$

Then

$$\mathrm{OQ}^2 - \mathrm{MQ}^2 = \mathrm{b}^2\,\mathrm{cosec}^2\,\theta - \left(\mathrm{b}\,\mathrm{cosec}\,\theta - \frac{b\sin\theta}{\cos\theta+1}\right)^2$$
$$= \frac{2\,\mathrm{b}^2}{\cos\theta+1} - \frac{\mathrm{b}^2\sin^2\theta}{(\cos\theta+1)}$$
$$= \frac{\mathrm{b}^2}{\cos\theta+1}\left(\frac{2\cos\theta+2-\sin^2\theta}{(\cos\theta+1)}\right)$$
$$= \frac{\mathrm{b}^2}{\cos\theta+1}\left(\frac{2\cos\theta+1+\cos^2\theta}{(\cos\theta+1)}\right)$$
$$= \mathrm{b}^2$$
$$= 4$$

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